18/7/26

DS - Set 1

Scenario based

1) * Assign the values of the dataset wrt a variable called 'dataset'.

* Code as dataset.describe()

* 25% of the value is Q_1 value and 75% of value is ~~75%~~ Q_3 value.

* IQR can be found by following formula

$$IQR = Q_3 - Q_1$$

* Lower range and upper range can be found by following code

$$\text{Lower range} = Q_1 - 1.5 \times IQR$$

$$\text{Outlier greater} = Q_3 + 1.5 \times IQR$$

2) * Assign the data to a variable called 'dataset'.

* To find.

i) Mean - use the code dataset. mean(). item()

ii) Median - use the code dataset. median(). item()

iii) Mode - use the code dataset. mode()[0]. item()

*) Median will be a better option because if the outliers are present mean is affected but median is not. So median is good.

3) Frequency table :-

no. of Customers	Frequency
5	2
10	2
8	1
12	1
14	1
15	1
18	1
20	1

4) *) Assign the data to a dataset variable

*) To find multicollinearity

We have to calculate VIF [Variance Inflation Factor].

* If the vif > 5, it violates multicollinearity

5) * Assign the data to a variable called dataset.

* We need to assume H_0 and H_1.

* H_0 - Null hypothesis
  H_1 - Alternate hypothesis.

* The statement as follows:-
  H_0 - The medicine doesn't work for blood pressure
  H_1 - The medicine works for blood pressure

* Perform T-test using the code.

  * If p-value < 0.05% we reject null hypothesis.

  * If p-value > 0.05% we fail to reject null hypothesis.

6) * Assign the values of data to a variable called dataset.

* Perfor the Interquartile range to detect outliers

* Use the following code as below:

```
Q1 = dataset ["columnName"].quartile (0.25)
Q3 = dataset ["columnName"].quartile (0.75)

IQR = Q3 - Q1

outliers = dataset [ (dataset ["columnName"] <
                (Q1 - 1.5 * IQR) |
             (dataset ["columnName"] >
                (Q3 + 1.5 * IQR))]

print (outliers)
```

7) * Assign the data to a dataset variable.

* We need to find mode to know the overall rating most people given.

* Perform mean and median for addititonal information.

* It gives the idea of the customer satisfaction.